

Guerby Mertil

Senior Software Engineer

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Professional Summary

Senior Software Engineer with **12+** years of experience building and scaling **cloud-native, backend-centric systems**. Expert in **Java** and **Spring Boot**, with strong hands-on experience in **Node.js, Python**, and the **MERN stack**, delivering **microservices** and **event-driven** architectures at scale. Proven track record integrating **AI/ML** pipelines and inference services into production platforms to enable **data-driven personalization** and **automation**. Experienced in supporting **web, mobile**, and **Android** (Kotlin/Java) clients through well-designed, high-performance APIs. Known for technical **leadership, architectural** ownership, and **mentoring** engineers across complex, regulated domains including fintech, betting, and healthcare.

Technical Skills

Backend: Java 8–21, Java SE, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Spring Cloud, Spring WebFlux, Node.js, Express.js. Microservices, Event-Driven Architecture, REST API & API Design, gRPC, GraphQL, Kafka, RabbitMQ, Redis Streams, Concurrency & Multithreading, Domain-Driven Design (DDD), Clean Architecture, CQRS, Circuit Breakers, Resilience4j, Python

- **Frontend:** React.js, Next.js, Angular, React Native, TypeScript, JavaScript (ES6+), Redux Toolkit, React Query, NgRx, Context API, Component Architecture, Responsive UI, WebSockets, SSR/ISR, Mobile-first UI, Storybook
- **AI, ML & Data:** Machine Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Image Classification, Object Detection, Video Analytics, Feature Engineering, Model Training, Model Fine-Tuning, Model Inference, Prompt Engineering, Retrieval-Augmented Generation (RAG), Multimodal Models, Anomaly Detection, Representation Learning, Model Evaluation, Transfer Learning, Responsible AI
- **Mobile/Android:** Kotlin, Java(Android), Android Studio, Coroutines, MVVM, Material Design, Mobile API Integration, Mobile Debugging & Troubleshooting, App Lifecycle Management
- **Cloud & DevOps:** AWS (EC2, S3, ECS, EKS, RDS, DynamoDB, Lambda, API Gateway, CloudWatch, CloudTrail), Azure DevOps, GCP basics, Docker, Kubernetes, Terraform, Helm, Jenkins, GitHub Actions, GitLab CI, ArgoCD, Linux, Nginx, Infrastructure as Code, Secrets Management, Service Mesh (Istio/Linkerd)
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Oracle, SQLite, DynamoDB, Elasticsearch, Data Modeling, Indexing, Query Optimization, Sharding & Replication
- **Security & Compliance:** OAuth2, OpenID Connect, JWT, SAML, Spring Security Advanced, Keycloak, Vault, API Gateway Security, Rate Limiting, RBAC/ABAC, PCI-DSS (fintech), HIPAA concepts (healthcare), OWASP, Threat Modeling, Data Encryption (at rest & in transit)
- **Monitoring & Observability:** Datadog, Prometheus, Grafana, OpenTelemetry, ELK Stack, Jaeger, Zipkin, Spring Actuator, Distributed Tracing, Metrics & Alerting, Chaos Testing
- **Testing:** JUnit 5, Mockito, Testcontainers, WireMock, Pact (Contract Testing), Integration Testing, Load/Stress Testing (JMeter, k6), Jest, React Testing Library, Cypress, Test-Driven Development

Professional Experience

Senior Software Engineer — [Fanatics](#)

09/2023 – Present | Jacksonville, Florida

- Architected and delivered a **cloud-native, event-driven commerce platform** using **Java** and **Spring Boot** as the core backend, supporting **championship-level traffic** with consistent system stability.
- Led the redesign of a **legacy monolith** into **modular Java microservices**, enabling **independent scaling, faster deployments**, and improved **fault isolation** across teams.
- Designed **high-throughput backend services** processing **real-time commerce and event data**, reducing **end-to-end latency by up to 80%** during peak live events.
- Implemented **resilience patterns** across critical services, including **circuit breakers, retries, rate limiting**, and **distributed caching**, significantly reducing **production incidents**.
- Built **secure, versioned REST APIs** and **backend gateways** consumed by **web and mobile clients**, enforcing **enterprise authentication and authorization** standards.
- Developed **Node.js services** to complement **Java microservices**, enabling **flexible full-stack workflows** and faster iteration on **frontend-driven features**.
- Built and maintained **modern React applications** using **TypeScript** and current libraries, delivering **real-time dashboards** for inventory, manufacturing, and **live commerce visibility**.
- Improved **frontend performance and reliability** by optimizing **component architecture, state management**, and **API integration patterns** in **React** and **TypeScript**.
- Implemented **end-to-end full-stack features** by coordinating **Java backends, Node.js services**, and **React frontends**, reducing **handoff friction** between teams.
- Designed **real-time data update flows** for frontend applications, enabling **live operational insights** during **high-traffic events**.
- **Developed and deployed machine learning models** to generate personalized product recommendations and predictive insights. Leveraged **AWS SageMaker** and **Snowflake** to build scalable pipelines that improved **customer engagement** and helped drive e-commerce growth.
- **Integrated machine learning inference** into **Java backend services** to provide dynamic product recommendations via APIs, supporting **personalization** across our global e-commerce platforms.
- Designed and optimized **data ingestion and feature pipelines** to process event streams and transactional data, ensuring **production-grade inputs** for machine learning models. This data foundation powered core ML systems and supported **real-time customer insights**.
- Led the development of **LLM-powered betting intelligence** features that translated real-time **odds** movements, injury updates, and market shifts into clear, **natural-language** explanations, increasing **live-betting** engagement and reducing customer support inquiries related to bet comprehension.
- Contributed to **sports betting platforms** by developing predictive models for risk analysis and **trend forecasting**, ensuring a competitive edge in the rapidly evolving market.
- Enhanced **internal operational efficiency** by automating data processes and optimizing inventory planning workflows using AI-driven insights, improving accuracy and reducing manual intervention.
- Focused on **product personalization**, improving **customer experience** through sophisticated recommendation engines and personalized product ranking systems that increased retention rates and user satisfaction.

- Built **LLM-assisted risk and compliance** tooling to summarize **anomalous** betting patterns and user behavior at scale, enabling faster **investigation** by trust and safety teams while supporting responsible gaming initiatives and maintaining regulatory alignment in a **high-volume** wagering environment.
- Mentored engineers, participated in **architectural reviews**, and helped define **best practices** for **backend, full-stack, and AI-enabled systems**.

Senior Software Engineer — [EchoStar Corporation](#)

04/2020 – 04/2023 | Englewood, Colorado

- Designed and delivered **high-throughput backend microservices** using **Java** and **Spring Boot**, powering **large-scale satellite and OTT data platforms** with strict **reliability** and **latency** requirements.
- Led **end-to-end technical initiatives** from **architecture and planning** through **production delivery**, ensuring **highly available backend systems** aligned with **business and platform goals**.
- Provided **technical leadership and mentorship**, setting **engineering standards** across **backend, MERN-based frontend integrations, and distributed system design**.
- Actively contributed to **system architecture and design reviews** spanning **backend services, MERN-based applications, and mobile platforms**, strengthening **security, scalability, and long-term maintainability**.
- Conducted **in-depth code reviews**, managed **technical debt**, and implemented **automated CI/CD pipelines**, accelerating **deployment cycles** and improving **release stability**.
- Developed **backend services** supporting **multimedia networking and communication systems**, including **wireless and satellite data flows** consumed by **web and MERN-based client applications**.
- Integrated **AWS cloud infrastructure, Docker, Kubernetes, and Rancher** to support **scalable backend services and data pipelines** used for **analytics and AI-driven insights**.
- Designed and implemented **event-driven microservices** enabling **real-time data ingestion and processing**, forming the foundation for **downstream analytics and machine learning workflows**.
- Optimized **PostgreSQL, MySQL, and MongoDB** schemas and queries, improving **read/write performance by ~30%** and supporting **data-intensive AI and analytics use cases**.
- Implemented **distributed caching and messaging systems (Kafka, RabbitMQ, Redis)** to enable **low-latency, high-volume data flows** required by **real-time applications and ML pipelines**.
- Built **secure REST and gRPC APIs** consumed by **MERN-stack applications and internal AI services**, ensuring **reliable and scalable data exchange**.
- Developed and deployed **computer vision** pipelines to analyze **satellite imagery and infrastructure** visuals, enabling automated **detection** of anomalies, signal obstructions, and physical degradation that previously required **manual inspection** by operations teams.
- Designed high-performance **image and video** processing systems to extract **actionable insights** from **large-scale visual data streams**, improving monitoring accuracy and supporting faster **operational decisions** across **satellite and broadband** network environments.
- **Developed scalable ML models** using **AWS SageMaker and Snowflake** to generate personalized recommendations, optimizing user engagement and driving operational efficiencies in a high-traffic environment.
- **Engineered production-grade data pipelines**, leveraging **AWS services (S3, Lambda)**, ensuring reliable and efficient data flows for real-time AI insights. Applied **MLOps** practices, including **CI/CD pipelines**, to maintain model performance in live environments.
- **Collaborated with cross-functional teams** to develop AI-driven workflows for optimizing **operational processes**, such as inventory planning, with machine learning insights to improve **accuracy and efficiency**, aligning with **EchoStar's focus on operational optimization**.

Senior Software Engineer — [Thrive Market](#)

06/2015 – 03/2020 | Los Angeles, California

- Developed **Java microservices** for dealer scoring and CRM analytics, reducing **system response time by 50%** and improving **member satisfaction**.
- Supported Android and React Native client integrations through API improvements and performance optimization.
- Transformed legacy **Java applications** into modular **Spring Boot microservices**, enhancing **system reliability, maintainability, and scalability**.
- Designed and implemented **Java pricing microservices** for processing large volumes of daily **ecommerce transactions**.
- Built **REST and gRPC APIs** to support secure, reliable communication between microservices and frontend platforms.
- Implemented **distributed caching and message queues** to optimize real-time data processing and reduce latency.
- Developed **React Native fleet management tools** to track and optimize logistics, achieving an **18% reduction in operational costs**.
- Delivered **ecommerce services** including order editing, shipment splitting, and inventory synchronization, streamlining workflows and improving **operational accuracy**.
- Built responsive **React and Next.js components** for member dashboards and catalog management, improving **user engagement**.
- Integrated backend services with **cloud infrastructure (AWS EC2, S3, RDS, Lambda)** to support scalable and fault-tolerant operations.
- Conducted **code reviews, guided junior engineers**, and enforced **best practices** for backend and frontend development.
- Implemented **monitoring and observability** using Datadog and Prometheus, improving **system reliability and incident response**.
- Collaborated with **cross-functional teams** in Agile environments to deliver high-quality features on schedule.

Software Engineer — [Ensora Health](#)

03/2012 – 01/2015 | Birmingham, Alabama

- Developed **Java-based REST APIs** for **claims eligibility and enrollment**, supporting **100,000+ users annually** in healthcare systems.
- Optimized **multithreaded Spring Batch jobs**, reducing **claims processing time by 25%** and improving operational efficiency.
- Implemented **CI/CD pipelines** using **GitHub Actions and GitLab**, reducing **deployment failures by 50%** and accelerating delivery cycles.
- Conducted **monthly security audits** and enforced **HIPAA-compliant encryption protocols** for sensitive patient and claims data.
- Applied **Test-Driven Development (TDD)** with **JUnit and Mockito**, increasing **code coverage** and reducing **production defects by 40%**.
- Implemented secure mobile-backend integrations (OAuth2, encrypted storage, network hardening), improving application reliability and meeting rigorous best-practice standards.
- Integrated **cloud services (AWS S3, Lambda, RDS)** to enable **scalable, fault-tolerant, and secure healthcare solutions**.
- Refactored **legacy Spring Boot services**, enhancing **performance, maintainability, and scalability** of critical healthcare systems.
- Mentored **junior engineers** and conducted **code reviews**, promoting **best practices, secure coding, and high-quality backend development** in the healthcare domain.

Education

[West Virginia State University](#), Bachelor of Computer Science, 09/2000 – 05/2005 | Institute, WV

[Florida International Communication](#), 09/2002 – 06/2004 | Miami, FL